

SOE MIN THANT

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SUMMARY

Full-stack software engineer with experience building web applications and software platforms. Currently developing tools and applications for human-AI collaboration at SMART, with interests in software architecture and developer tools.

EDUCATION

National University of Singapore **Feb 2023 - May 2024**
Graduate Diploma in Systems Analysis

University of Gloucestershire **Jul 2016 - Jul 2020**
Bachelor of Science in Computing with Honours

WORK EXPERIENCE

Research Software Development Engineer **Jul 2024 - Present**
Singapore-MIT Alliance for Research and Technology (SMART) Singapore

- Built and maintained a full-stack ontology platform for modeling and exploring complex work processes using Next.js, TypeScript, and Firebase.
- Implemented real-time collaborative editing features for concurrent multi-user editing and live synchronization.
- Developed software tools and applications for human-AI collaboration research and workflow exploration.
- Designed and optimized data models and application workflows to support large-scale ontology management and visualization.
- Integrated LLMs to support content generation, workflow analysis, and AI-assisted decision support.
- Presented the platform at workshops and symposiums in Singapore, engaging with researchers, industry partners, and external collaborators.

Software Engineer Intern **Sep 2023 - Jan 2024**
Saltmine Singapore

- Contributed to full-stack development in a MERN-based microservice architecture.
- Refactored a monolithic backend service into a more maintainable service structure for evolving product requirements.
- Designed relational schemas and migration pipelines to transfer large datasets from Elasticsearch to PostgreSQL with a focus on data integrity.

Junior Software Developer **Apr 2021 - Mar 2022**
Seattle Consulting Myanmar

- Developed and maintained web applications using Ruby on Rails and Vue.js for human resources and employee management systems.
- Built internal tools for business operations and administrative workflows.

SKILLS

- **Languages:** TypeScript, JavaScript, Java, C#, Python
- **Frontend:** React, Next.js, Tailwind CSS, HTML5, CSS
- **Backend:** Spring Boot, Node.js, ASP.NET, Firebase
- **Databases:** PostgreSQL, MongoDB, Firestore, MySQL, Elasticsearch, Redis
- **Tools & Technologies:** Git, Docker, Jenkins, ArgoCD, Jest, Cypress, LLM Integration

PROJECTS

Ontology of Collective Intelligence (SMART)

[Website](#) | [GitHub](#)

- Developed a large-scale ontology platform (knowledge base system) with hundreds of thousands of entities, supporting collaborative editing and exploration for researchers and external industry partners.
- Implemented real-time collaborative editing with Yjs, including conflict-safe change persistence and server-side edit history tracking for concurrent users.
- Designed and optimized hierarchical and graph-based data models in Firestore, resolving document size limits and performance bottlenecks through data partitioning and on-demand loading.

Activity Logging System (Saltmine)

- Migrated a high-traffic activity logging system from Elasticsearch to PostgreSQL to improve data integrity, maintainability, and scalability.
- Designed a normalized relational schema and data transformation pipelines to preserve large volumes of historical activity data.
- Refactored related backend services from Express.js to NestJS to improve separation of concerns from other services and long-term maintainability.

CodeMap - Codebase Understanding Tool (SMART)

- Contributed to CodeMap, an LLM-assisted codebase understanding tool that helps developers explore large codebases through interactive visualizations.
- Built frontend interaction flows for navigating code structures across multiple abstraction levels.
- Integrated LLM-based queries to support context-aware code analysis and user-driven exploration.

SwiftBackend - Backend-as-a-Service Prototype (Personal)

- Designed and prototyped a multi-tenant backend platform for API generation, authentication, and workflow automation.
- Built backend services using Spring Boot, Go, and TypeScript, exploring tenant isolation, schema management, and secure database provisioning.
- Developed a custom API gateway for authentication, routing, rate limiting, and service orchestration.

PUBLICATIONS & AWARDS

- Co-author, “Understanding Codebase like a Professional! Human-AI Collaboration for Code Comprehension,” IEEE/ACM ICPC 2026.
- HND Excellence Award 3rd Prize, 2017